

AIML Capstone 3: Sales Forecasting

Fresh Analytics forecasts restaurant item demand using merged sales data, EDA, and ML models.

1. Business context

Accurate demand forecasts drive production, staffing, and inventory decisions across restaurant chains.

2. Data preparation

Sources. restaurants.csv (6 stores), items.csv (100 items), sales.csv (109,600 daily records).

Merge. Single table with date, item, price, count, kcal, store id/name.

Cleaning. Outlier cap at 99.5th percentile on item_count (241 units).

3. Exploratory analysis

Daily/weekly/monthly/quarterly sales trends; day-of-week patterns.

Bob's Diner leads in volume and revenue; per-store and per-item breakdowns.

Most expensive items and calorie counts identified per restaurant.

4. Forecasting models

Features. Year, month, day, day-of-week, quarter, lag/rolling stats.

Split. Last 6 months as test (train 88,200 / test 18,400 rows).

RMSE (test). Linear Regression 3.16; Random Forest 3.07; XGBoost 2.88 (best).

Forecast. XGBoost used for iterative one-year (2022) forward forecast.

5. Tools & conclusion

pandas, scikit-learn, XGBoost, matplotlib. XGBoost generalises best for this demand series.